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Scalable network monitoring platform

Abstract

A scalable monitoring platform for use with a target network having a network management device and multiple managed devices. The platform includes a remote monitoring server and a monitoring device co-located with the target network. The monitoring device is configured to communicate directly with the managed devices using a network management protocol, and to retrieve management data using separately generated and transmitted targeted requests to each managed device. Each managed device transmits a response and each response is evaluated to identify changes in host status or service status indicative of an adverse condition. Dynamic monitoring policies are implemented in response to the changes and management actions are executed in response to adverse conditions.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5049873	12/1990	Robins	N/A	N/A
6061725	12/1999	Schwaller	N/A	N/A
7003564	12/2005	Greuel	709/224	H04L 41/5009
7685316	12/2009	Sukumaran	709/218	H04L 41/0843
8635319	12/2013	Cuthbert	N/A	N/A
10003493	12/2017	Kim	N/A	N/A
10979335	12/2020	Li	N/A	N/A
11258653	12/2021	Waldbusser	N/A	H04L 43/0817
2011/0302652	12/2010	Westerfeld	N/A	N/A
2015/0081891	12/2014	Baba	709/224	G06F 21/105
2017/0187587	12/2016	Keppel	N/A	N/A
2019/0028363	12/2018	McBride	N/A	N/A
2019/0286500	12/2018	Tucker	N/A	G06F 9/542
2020/0393807	12/2019	Muranaka	N/A	G05B 19/052
2021/0194782	12/2020	Jump	N/A	N/A
2021/0273953	12/2020	Fellows	N/A	N/A
2021/0282224	12/2020	Kanitkar	N/A	N/A
2023/0379300	12/2022	Chundury	N/A	H04L 63/0236

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a nonprovisional utility application of provisional patent application, Ser. No. 63/414,611 filed in the United States Patent Office on Oct. 10, 2022, claims priority therefrom, and is expressly incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to a system for monitoring a network to detect adverse conditions. More particularly, the present disclosure relates to a scalable monitoring platform utilizing targeted requests directed to individual network elements to achieve rapid and accurate detection of adverse conditions.

BACKGROUND

(2) Network management systems play a critical role in monitoring the health of computer networks, by detecting faults and errors affecting a network, assessing the operational state of

individual devices within the network, and collecting performance data related to tasks carried out individually by the managed devices. Typically, conventional management systems retrieve management data through mass polling of all the managed devices within the network, and produce a single aggregated data output containing the data returned by every managed device. The aggregated data output is collected and stored by a single device or computer on the network acting as a manager. This aggregated data output may be referred to as a data dump.

(3) The conventional approach is hampered by several key drawbacks which increase in severity as networks grow in scale and complexity. When a manager polls one of the managed devices, the managed device will react by transmitting all of the data that it is configured to collect, resulting in the transmittal and recording of much irrelevant data. Furthermore, conventional management systems will poll every managed device on the network in order to produce the aggregated data output, often in a sequential fashion hampered by delayed responses, timeout errors, and general network congestion. In certain cases, timeout errors may cause the manager to repeat the entire polling process, thus potentially creating frequent gaps in the collected data in which adverse conditions or events go undetected. In addition, the lengthy amount of time required to complete the polling process produces data which may be stale by the time analysis and review can begin.

(4) Furthermore, expanding the number of managed devices will increase the size of the aggregated data output, thus increasing the cost of analysis as computing resources are devoted to reviewing and processing ever-increasing amounts of irrelevant data in order to spot events or identify trends.

(5) It is therefore clear that an urgent need exists for a monitoring platform which overcomes obstacles to scalability through targeted requests directed to individual managed devices for specific management data items. Such requests will be generated independently and transmitted in a simultaneous manner, thus avoiding the delays and loss of data incurred using conventional polling techniques. Furthermore, each targeted request utilizes management commands native to the network management protocol of network, and causes the managed device to reply by promptly transmitting a response containing only the management data items specified by the request. Such responses will be evaluated upon receipt using algorithms specific to the managed device, resulting in rapid and accurate evaluation of the management data.

(6) In the present disclosure, where a document, act or item of knowledge is referred to or discussed, this reference or discussion is not an admission that the document, act or item of knowledge or any combination thereof was at the priority date, publicly available, known to the public, part of common general knowledge or otherwise constitutes prior art under the applicable statutory provisions, or is known to be relevant to an attempt to solve any problem with which the present disclosure is concerned.

(7) While certain aspects of conventional technologies have been discussed to facilitate the present disclosure, no technical aspects are disclaimed and it is contemplated that the claims may encompass one or more of the conventional technical aspects discussed herein.

BRIEF SUMMARY

(8) An aspect of an example embodiment in the present disclosure is to provide a scalable platform capable of monitoring any number of devices within a target network comprising a plurality of managed devices and a network management device. The network management device and the managed devices are adapted to communicate via a network management protocol. Each managed device is configured to provide management data, including host data describing functional conditions of the managed device, and service data describing a service task which the managed device is configured to carry out. Accordingly, the present disclosure provides a scalable monitoring platform for monitoring the target network, comprising a monitoring device and a remote monitoring server. The monitoring device is co-located with and communicates directly with the target network, and is configured to request a management data item from each managed device by transmitting a targeted request directed to each managed device. The targeted requests are generated and transmitted separately or in parallel using management commands native to the

network management protocol, and incorporate address data obtained via the network management protocol allowing precise targeting of managed devices. Each managed device replies independently to the targeted request by transmitting the requested management data item. By transmitting targeted requests for individual data items separately at different times or in a synchronous, parallel manner, the scalable monitoring platform avoids bottlenecks and timeouts which hamper scalability of conventional monitoring systems.

(9) It is another aspect of an example embodiment in the present disclosure to provide a scalable platform capable of rapidly detecting adverse conditions related to the functional condition or service task condition of individual managed devices. Accordingly, the scalable monitoring platform utilizes a data evaluation module configured to assess each management data item separately using evaluation criteria specific to the managed device. Targeted requests are directed to individual items of host data or service data from the managed device, and the data evaluation module evaluates each data item upon receipt. This allows the data evaluation module to detect changes to the functional condition or service task condition of each managed device without delays caused by transmission and evaluation of irrelevant data. Furthermore, the evaluation criteria contains a plurality of ranges which allow the data evaluation module to assess severity of adverse conditions indicated by the management data.

(10) It is yet another aspect of an example embodiment in the present disclosure to provide a scalable platform capable of dynamically scheduling requests in reaction to changes in functional condition or service task condition of the managed devices. Accordingly, the scalable monitoring platform is configured to execute monitoring policies for each managed device, and each monitoring policy has a standard operating mode and one or more adverse operating modes. If the data evaluation module detects an adverse changes to the functional condition or service condition of a managed device, the monitoring policy for the managed device switches to one of the adverse operating modes. The adverse operating mode causes the request module to generate additional targeted requests at increased frequency to verify the adverse condition. Furthermore the scalable monitoring platform is adapted to initiate management actions to notify users of adverse conditions, and initiate mitigating device-level actions by the managed devices, such as reboots.

(11) It is contemplated that the present disclosure may prove useful in addressing other problems and deficiencies in a number of technical areas. Therefore, the claims should not necessarily be construed as limited to addressing any of the particular problems or deficiencies discussed hereinabove. To the accomplishment of the above, this disclosure may be embodied in the form illustrated in the accompanying drawings. Attention is called to the fact, however, that the drawings are illustrative only. Variations are contemplated as being part of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, like elements are depicted by like reference numerals. The drawings are briefly described as follows.

(2) FIG. 1A is block diagram depicting a remote monitoring platform and a target network with a network management device and plurality of managed devices, in accordance with an embodiment in the present disclosure.

(3) FIG. 1B is a block diagram depicting a plurality of targeted requests being transmitted to the managed devices, as well as responses containing management data transmitted in reply by the managed devices, in accordance with an embodiment in the present disclosure.

(4) FIG. 1C is block diagram depicting an example remote monitoring server, in accordance with an embodiment in the present disclosure.

(5) FIG. 1D is block diagram depicting an example managed device and subcomponents, in

accordance with an embodiment in the present disclosure.

(6) FIG. 2A is block diagram showing a network mapping module retrieving network configuration data from the target network, in accordance with an embodiment in the present disclosure.

(7) FIG. 2B is block diagram showing an example network map of the target network, and managed device data which allows the remote monitoring platform to locate and communicate with each manage device, in accordance with an embodiment in the present disclosure.

(8) FIG. 3A is a block diagram showing an example set of managed device data and evaluation criteria, in accordance with an embodiment in the present disclosure.

(9) FIG. 3B is a block diagram showing targeted requests being generated using request parameters determined by a monitoring policy, in accordance with an embodiment in the present disclosure.

(10) FIG. 4 is block diagram showing response data being passed to a data evaluation module, in accordance with an embodiment in the present disclosure.

(11) FIG. 5 is a block diagram depicting a data conversion function and data evaluation function of the data evaluation module, in accordance with an embodiment in the present disclosure.

(12) FIG. 6 is a flowchart depicting an example data evaluation process, in accordance with an embodiment in the present disclosure.

(13) FIG. 7 is a block diagram showing management actions being carried out by a notification module, further showing external action commands transmitted to an external management system, in accordance with an embodiment in the present disclosure.

(14) The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, which show various example embodiments. However, the present disclosure may be embodied in many different forms and should not be construed as limited to the example embodiments set forth herein. Rather, these example embodiments are provided so that the present disclosure is thorough, complete and fully conveys the scope of the present disclosure to those skilled in the art.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(15) FIGS. 1A-B illustrate a scalable monitoring platform **10** for managing one or more target networks **20**. In a preferred embodiment, the scalable monitoring platform **10** comprises one or more monitoring devices **14** locally positioned in relation to one of the target networks **20**, one or more remote monitoring servers **12** configured to communicate with the monitoring devices **14**, and at least one platform data store **30**. The scalable monitoring platform **10** may also comprise one or more user devices **16** and a platform interface **18** which allow a user to interact with the scalable monitoring platform **10**. Each user device **16** may correspond to a personal computer, mobile phone, or other computing device capable of accepting user commands, displaying text and graphics within a graphical user interface, and communication via the internet **500**. The platform interface **18** may be implemented as an application **18A** on the user device **16** or within a platform web page **18W**. In some embodiments, the scalable monitoring platform **10** is further configured to communicate with an external management system **100**.

(16) Each target network **20** corresponds to an open or closed network comprising a network management device **22** and a plurality of managed devices **24**, whereby the network management device **22** and the managed devices **24** communicate using a network management protocol **45**. The network management protocol **45** may be implemented at the application layer to provide a common set of management commands and database schema for storing and using management data **58**.

(17) In a preferred embodiment, the monitoring device **14** is configured to communicate directly with the target network **20** using the network management protocol **45**. In addition, the monitoring device **14** is configured to communicate with the remote monitoring server **12** and exchange platform data via a wide area network **500** (WAN) such as the Internet. In certain embodiments, the target network **20** may be deployed within a network site corresponding to a geographic location, such as a building or portion thereof, an outdoor area, or other location with defined boundaries.

The monitoring device **14** may therefore be deployed within the network site.

(18) The network management device **22** is operably linked to the managed devices **24** within the target network **20**, and is adapted to transmit management commands to each of the managed devices **24** using the network management protocol **45**. Each of the managed devices **24** is adapted to respond to such management commands by transmitting a response **70** containing management data **58**. In a preferred embodiment, each managed device **24** is configured as a host within the target network **20**, and is adapted to perform one or more service tasks.

(19) The scalable monitoring platform **10** is configured to retrieve management data **58** from the target network **20** by sending targeted requests **52** to each of the managed devices of the target network **20** via the network management device **22**. The network management device **22** will promptly relay the targeted request **52** to the managed device **24**. In one embodiment, these targeted requests **52** are transmitted from the remote monitoring server **12** to the network management device **22** through the monitoring device **14**. The targeted request **52** is then relayed to the managed device **24** by the network management device **22**. In other embodiments, the targeted request **52** may instead originate from the monitoring device **14**.

(20) Each targeted request **52** may be addressed to a single managed device **24**, and may specify one or more items of management data **58**. The management data **58** may comprise host data **58A**, service data **58B**, or a combination thereof. The host data **58A** provides information regarding one or more functional conditions of the managed device **24**, while the service data **58B** provides information regarding the service tasks performed by the managed device **24**. For example, the host data **58A** may indicate whether the managed device **24** is operating normally and is able to communicate with the target network **20**, while the service data **58B** is used to monitor and evaluate the service task condition. Upon receiving the targeted request **52**, the managed device **24** will reply with a response **70** containing the requested management data **58**. The response **70** will be transmitted by the network management device **22** to the monitoring device **14**, and will then be relayed to the remote monitoring server **12** for evaluation.

(21) Each targeted request **52** is transmitted independently, and each managed device **24** will reply with a response **70** separately from other managed devices **24** within the target network **20**. Furthermore, the remote monitoring platform **10** may transmit any number of targeted requests **52** to any number of managed devices **24** within the target network **20**, limited by only by network and processing capacity of the target network **20**, monitoring device **14**, and the network management device **22**.

(22) Note that in an alternate embodiment, certain network management protocols **45** may allow the monitoring device **14** to transmit targeted requests **52** directly to managed devices **24** within the target network **20** without relaying the request through a network management device **22**. In such an alternative embodiment, the targeted requests **52** and resulting responses **70** may be exchanged directly between the monitoring device **14** and the corresponding managed devices **24**.

(23) In a preferred embodiment, the scalable monitoring platform **10** is configured to monitor each managed device **24** in accordance to one or more monitoring policies **50**. The management data **58** contained within the response **70** transmitted by each managed device **24** is evaluated, and the monitoring policy **50** may be adjusted if trends, degradation, or adverse conditions are identified as a result of the evaluation. Furthermore, the remote monitoring platform **10** maintains a data log **78** within the platform data store **30** for each target network **20** which records the values of the management data **58** received from each managed device **24**. The data log **68** may correspond to a time series database containing the value of each instance of management data **58** along with timing information and metadata to facilitate visualization and analysis.

(24) Referring to FIG. 2B while also referring to FIGS. 1A-B, the network management device **22** and the managed devices **24** each correspond to one of a plurality of nodes **26** within the target network **20**, and the physical and logical relationship of the nodes **26** may be described using a network topology or network map. In certain embodiments, the network management device **22**

corresponds to a gateway node **26G** while the managed devices **24** each correspond to a network node **26** within the target network **20**. The managed devices **24** are disposed downstream of the network management device **22**, and the network management device **22** controls access to each of the managed devices **24** from sources which are external to the target network **20**.

(25) In one embodiment, the target network **20** is configured as a localized network in which the network management device **22** and the managed devices **24** are operably linked via a wired medium or via a wireless communication protocol. Each managed device **24** corresponds to an electronic device which is capable of communicating over the target network **20** with other managed devices and the network management device **22**. In one purely illustrative example, four managed devices **24** are configured downstream of the network management device **22** acting as the gateway node **26G**. The managed devices **24** include a router **24R** configured as an intermediate node **26N** downstream of the gateway node **26G**, with three other managed devices **24** configured as end nodes **26E** downstream of the router **24R**.

(26) In one embodiment, the monitoring device **14** is directly connected to the target network **20** via the network management device **22**. To facilitate a direct wired or wireless connection between the monitoring device **14** and the target network **20**, the monitoring device **14** may be deployed on-site with target network **20**, thus allowing the monitoring device **14** to communicate with the network management device **22** via the network management protocol **45** of the target network **20**. The monitoring device **14** is an edge device simultaneously configured to communicate with the remote monitoring server **12** via the wide area network **500**.

(27) Each managed device **24** may correspond to a personal computer, terminal, switch, router, server, sensor, camera, antenna-equipped transmission device, or other networked device, while the network management device **22** corresponds to a computing device configured to transmit a query for electronic data from the managed devices **24** via the management commands provided by the network management protocol **45**. In one example, the managed devices **24** include a plurality of antennae-equipped devices which allow the targeted network **20** to function as a Distributed Antenna System (DAS) for providing wireless network access within a building or other geographic location. In yet another example, the managed devices **24** include a plurality of moisture sensors distributed at physical locations throughout a golf course.

(28) The host data **58A** returned by each of the managed devices **24** may be used to indicate whether the managed device **24** is in operational or inoperable functional condition. For example, the host data **58A** may represent a variety of information related to the operational status of the managed device **24**, such as power, network connectivity, battery, and other indicators as appropriate. The service data **58B** is used to quantify one or more service tasks associated with the managed device **24**. For example, the service data **58B** may include moisture reading data from moisture sensors embedded within the soil of the golf course, or signal strength or network traffic data readings transmitted by the DAS units.

(29) In one embodiment, the network management protocol **45** may be implemented as a version of Simple Network Management Protocol (SNMP), and the network management device **22** corresponds to a network management station (NMS) operating an SNMP Manager application. Each of the monitored devices **24** may be configured to operate an SNMP Agent application which allows the managed device **24** to receive and respond to data queries transmitted by the network management device **22** using the management commands provided through the network management protocol **45**. Note that the features of the remote monitoring platform **10** can be applied using alternate management protocols as will be apparent to a person of ordinary skill in the art in the field of the invention.

(30) Turning to FIGS. **1C-D** while also referring to FIGS. **1A-B**, the remote monitoring server **12** is adapted to execute a plurality of platform functions which allow the scalable monitoring platform **10** to communicate with and monitor multiple target networks **20**. The remote monitoring server **12** also allows each of the monitoring devices **14** to be commanded or configured remotely. The

remote monitoring server **12** is configured as a computing device having a processor **300**, a RAM **302**, a ROM **303**, a computer storage device **304**, and a communication module **305**. The communication module **305** is configured to communicate with the monitoring devices **14** and user devices **16** via the WAN **500**. In a preferred embodiment, the remote monitoring server **12** may be implemented using server hardware with sufficient processing power to carry out the necessary platform functions. In certain embodiments, the functions of the remote monitoring server **12** may be distributed or duplicated across multiple computing devices. The platform data store **30** may be maintained within the computer storage device **304** of the remote monitoring server **12**, or within a computer storage device **304** contained within a separate computing device which is accessible to the remote monitoring server **12**. Note that in certain embodiments, the monitoring devices **14** may be configured as computing devices with capabilities similar to the remote monitoring server **12**, allowing certain platform functions to be executed locally by the monitoring device **14** at the network site of the target network **20**.

(31) Referring to FIGS. 2A-B and FIG. 5 while also referring to FIGS. 1A-C, the remote monitoring server **12** has a plurality of platform modules adapted to carry out the various platform functions. These platform modules may be implemented using software components, packages, or assemblies, as will be apparent to a person of ordinary skill in the art in the field of the invention. In one embodiment, the modules comprise a network mapping module **42**, a request module **44**, a data evaluation module **46**, and a notification module **90**. In one embodiment, the network mapping module **42** is adapted to identify the active network management protocol **45** used by the target network **20**, retrieve target network configuration data **32** from the target network **20**, and define a managed device list **24L** which references the managed devices **24** present within the target network **20**. The request module **44** is adapted to generate targeted requests **52** which are compatible with the network management protocol **45**, in accordance with the monitoring policies **50**. The data evaluation module **46** is adapted to interpret and evaluate the management data **58** transmitted by each managed device **24**, and initiate a management action **102** in reaction to the evaluation of the management data **58**. Referring briefly to FIG. 7, in one embodiment, the management actions **102** may include displaying a platform alert **94** using the platform interface **18**, as well as transmitting an external action command **98** to an external management system **100**. Note that in certain embodiments, certain functions of the platform modules may be executed from the monitoring device **14**, while other functions are executed remotely by the remote monitoring server **12**.

(32) Referring to FIGS. 1C-D while also referring to FIGS. 1A-B and FIGS. 2A-B, each network management device **22** is a network capable computing device operably linked to each of the managed devices **24** of the target network **20**, and may be implemented using server-grade hardware capable of processing a high volume of data from a large number of managed devices **24**. The managed devices **24** may be implemented using various computing devices or network capable systems each configured to receive and respond to management commands using the network management protocol **45**. For example, a managed device **24** configured as a sensing unit **24S** may have a control module **306**, one or more subcomponents **25** (such as sensors), and a communication module **305** for communicating with other managed devices **24** and the network management device **22**. The control module **306** is configured to operate the subcomponents **25** and may also be configured to monitor the functional condition of the managed device **24**.

(33) Referring to FIGS. 2A-B, FIGS. 1A-B, and FIG. 1D, the network mapping and discovery module **42** is configured to retrieve the target network configuration data **32** of the target network **20**. In certain embodiments, the network management protocol **45** may provide a network discovery function which reveals the managed devices **24**, and which may also be used to identify the network topology or network map of the target network **20**. Furthermore, the target network configuration data **32** contains sufficient data to allow the remote monitoring server **12** to identify each of the managed devices **24** and generate a managed device list **24L** for the target network **20**.

In one embodiment, the remote monitoring server **12** may transmit a network discovery request **42R** to the target network **20** via the monitoring device **14**, whereupon the target network configuration data **32** is in turn relayed to remote management server **12**.

(34) In one example where SNMP is the active network management protocol **45**, the target network configuration data **32** is stored within a management database **28** maintained by the network management device **22**, and which contains one or more Management Information Base files (MIB) **28B**. Each MIB **28B** contains data which reveals the hierarchical order and arrangement of the managed devices **24** within the target network **20**. Each managed device **24** is referenced using a unique identifier, such as an Object Identifier (OID), which further serves as an address for the managed device **24** within the target network **20**. SNMP uses a standardized system of notation in which each OID comprises an ordered list of components. Each component within the OID may be used to identify a characteristic of the managed device **24**, such as the vendor or manufacturer, equipment type, as well as subcomponents **25** which are responsible to collecting particular items of management data **58**. The MIB **28B** may further define the data types and ranges of the management data **58** associated with the managed device **24**.

(35) The information present within the management database **28** associated with each managed device **24** is used to populate a set of managed device data **34** which will be used by the scalable monitoring platform **10** to communicate with the managed device **24**. The managed device data **34** may contain address data **36** which allows targeted requests **52** to be directed to the management device **22** using management commands compatible with the network management protocol **45** of the target network **20**. The managed device data **34** also defines one or more associated data items **38** of the managed device **24**, each corresponding to an item of host data **58A** or service data **58B**. The address data **36** may comprise a device network address **36A** which identifies or locates the managed device **24** within the target network **20**, and one or more device-level identifiers **36D** each associated with one of the associated data items **38**. In one embodiment, the device-level identifier **36D** may be combined with the device network address **36A** to reference a specific item of host data **58A** or service data **58B** of the managed device **24**. The address data **36** may be retrieved by locating and parsing the OID of the managed device **24** within the appropriate MIB file **28B**, and the device-level identifiers **36D** may correspond to particular components of the OID.

(36) The OID may also be used to store information such as a physical location linked to the managed device **24** or one of its associated data items **38**. For example, the physical location **38L** may correspond to a specific location within the network site at where the managed device **24** is physically installed, or a location which is monitored by one of the subcomponents **25** of the managed device **24**. Therefore, the managed device data **34** of a managed device **24** may be populated with a physical location descriptor **38L** using the information present within the OID and the MIB **28B**.

(37) Each associated data item **38** is linked to evaluation criteria **38E** which allows the management data **58** to be evaluated by the scalable monitoring platform **10**. In a preferred embodiment, the evaluation criteria **38E** linked to each associated data item **38** includes ranges or thresholds **38ET** against which the value of the host data **58A** or service data **58B** may be compared to determine the functional condition or service task condition as appropriate.

(38) Each associated data item **38** may further be linked to a data descriptor **38D** which describes the associated data item **38** to facilitate organization and ease of analysis. In certain embodiments, the MIB **28B** associated with each managed device **24** may be populated with labels which identify each of the OID components. In other embodiments, the platform data store **30** may be configured with a repository of known MIB configurations which allow the remote monitoring server **12** to identify the OID components of each managed device **24** and reference the appropriate data descriptor **38D** and evaluation criteria **38E**.

(39) The managed device data **34** may also contain authentication information **39** applicable to the target network **20** or the managed device **24**, such as encryption and login information. Such

authentication information **39** may be utilized by the request module **44** to generate the targeted request **52**.

(40) Turning to FIG. **3A** while also referring to FIG. **1B**, FIG. **1D**, and FIG. **2B**, an example set of managed device data **34** linked to a managed device **24** configured as a moisture sensor is shown. The address data **36** includes a device network address **36A** which allows the managed device **24** to be referenced on the target network **20**, along with a plurality of device level identifiers **36D** each linked to one of the associated data items **38** of the managed device **24**. In the present example, the managed device **24** is configured to monitor moisture and temperature at a physical location within the network site. One of the associated data items **38** is linked to a device level identifier **36D** which addresses a subcomponent **25** labeled as “Water Content Sensor”, and has a data descriptor **38D** of “Moisture Level”. A second associated data item **38** is linked to a device level identifier **36D** which addresses a subcomponent labeled as “Temperature Sensor”, and has a data descriptor **38D** of “Temperature Reading”. The associated data items of “Moisture Level” and “Temperature Reading” each correspond to service data **58B**, and are associated with evaluation criteria **38E**. In one example, the evaluation criteria **38E** may identify a range of data values for a plurality of service ranges comprising a standard range **38ES**, a warning range **38EW**, and a critical range **38EC**. The standard range **38ES** denotes data values which indicate the service task condition is in normal or acceptable condition, while the warning range **38EW** and the critical range **38EC** denote data values which indicate an adverse service task condition of increasing severity. Note that the evaluation criteria **38E** for each associated data item **38** may include more or fewer service ranges than illustrated in the present example.

(41) The managed device data **34** may further describe an associated data item **38** having a data descriptor of **38D** “Connectivity Data”, which is linked to a device level identifier **36D** which addresses a subcomponent **25** or communication module **305** labeled as “Network Adapter”. In the present example, the data descriptor **38D** of “Connectivity Data” corresponds to host data **58A** used to determine the functional condition of the managed device, and may be used to evaluate whether the managed device **24** is effectively communicating with the target network **20**. In one example, the evaluation criteria **38E** may correspond to a threshold value **38ET**.

(42) Multiple instances of functionally identical managed devices **24** may be distinguished by device network address **36A**, or by a unique identifier implemented as an associated data item **38**, such as a serial number.

(43) Turning to FIG. **3B** while also referring to FIG. **1A**, FIGS. **2A-B** and FIG. **3A**, the scalable monitoring platform **10** is configured with one or more monitoring policies **50** which are applicable to managed devices **24** within the managed device list **24L** of the target network **20**. Each monitoring policy **50** may be associated with one or more managed devices **24**, and each monitoring policy **50** may be configured with separate rules or evaluation criteria **38E** for monitoring and evaluating each of the associated data items **38** of the managed device **24**. For example, a monitoring policy **50** associated with a managed device **24** having three associated data items **38** may have distinct evaluation criteria **38E** for each of the three associated data items **38**.

(44) In one embodiment, each monitoring policy **50** is configured to cause the request module **44** to generate targeted requests **52** using a variety of request parameters **54**. In one embodiment, the request parameters **54** may comprise a target device **55D** corresponding to one of the managed devices **24**, a request type **55T** which can be formed as a host status request **55TA** or a service status request **55TB**, and a requested data item **55R** corresponding to one of the associated data items **38** of the targeted managed device **24**. The monitoring policy **50** may further define a request interval **62**, which governs timing and/or frequency of the targeted request **52**.

(45) Furthermore, each monitoring policy **50** may have a plurality of operating modes. In one embodiment, the operating modes may comprise a standard operating mode **50S**, and one or more adverse operating modes corresponding to a warning operating mode **50W** and a critical operating mode **50C**. In one embodiment, each monitoring policy **50** begins operation in the standard

operating mode **58S** by default. The scalable monitoring platform **10** utilizes the monitoring policies **50** to dynamically generate targeted requests **52** directed to the appropriate managed devices **24** in reaction to changes in functional conditions and service task conditions as expressed within the management data **58**. The request interval **62** may be shortened or lengthened depending on the current operating mode of the management policy **50**.

(46) Targeted requests **52** and any corresponding responses **70** are transmitted independently, and responses **70** are evaluated and recorded promptly upon receipt. Therefore, the quantity and frequency of targeted requests **52** do not increase the risk of data loss resulting from timeout errors. As such, the request intervals **62** may be shortened to minimize the risk of adverse conditions remaining undetected due to gaps in the retrieval, evaluation, and logging of management data **58**.

(47) The monitoring policy **50** associated with each managed device **24** may be predefined and stored within the platform data store **30**, and may further be customized or edited by a user via customization tools provided within the platform interface **18**. Such customizations and edits may be transmitted to the remote monitoring server **12**. The remote monitoring server **12** may then relay the modified monitoring policies to the monitoring device **14** co-located with the appropriate target network **20** for local implementation. In certain embodiments where the monitoring device **14** is configured to perform the functions of the request module **44**, the monitoring policies **50** associated with the target network **20** may be stored locally using the monitoring device **14**.

(48) Each monitoring policy **50** allows the scalable monitoring platform **10** to monitor and evaluate the host condition and service task condition of the managed device **24** through targeted requests **52** generated in accordance with the request interval **62**. The monitoring policy **50** may assign a separate request interval **62** for each associated data item **38** of the targeted managed device **24**, thereby allowing the targeted requests to **52** to be timed to reflect the type of the management data **58** being requested, the current functional condition of the managed device **24**, the nature of the service tasks performed by the managed device **24**, or any adverse events reflected in the management data **58**.

(49) In one example, a managed device **24** may be an outdoors moisture sensor configured to collect service data **58B** corresponding to localized moisture levels and temperature readings on a golf course. The monitoring policy **50** associated with this managed device **24** may be configured to schedule targeted requests **52** for service data **58A** at relatively long request intervals **62**, such as one request every five minutes, while requests for host data **58A** may be scheduled at relatively short request intervals **62**, such as one request every 30 seconds. In the present example, shifts in moisture levels and temperature occur relatively infrequently, while the functional condition of the managed device **24** itself must be regularly monitored. In certain embodiments, request intervals **62** may be linked to a particular series of times or time intervals to facilitate consistent and ordered logging of management data **58**. For example, all requests with a five minute request interval **62** may be generated at the times of “12:00, 12:05, 12:10 . . .”. Targeted requests **52** scheduled for the same time or time interval may therefore be generated and transmitted simultaneously in a synchronous manner.

(50) In a preferred embodiment, the monitoring policies **50** are protocol-agnostic, and a monitoring policy **50** configured for use with a particular managed device **24** may still be applied to multiple target networks **20** each implementing different network management protocols **45**. In one embodiment, the request module **44** utilizes a protocol compatibility submodule **44P** to translate the request parameters **54** into a format compatible with the target network **20** and its corresponding network management protocol **45**. The protocol compatibility submodule **44P** may be configured to match the network management protocol **45** utilized by the target network **45** to a plurality of known network management protocols **45**, populate the request parameters **54** using the managed device data **34**, and then combine the address data **36** with the appropriate management command native to the appropriate network management protocol **45**. The protocol compatibility submodule **44P** may be configured to select and utilize management commands from a range of network

management protocols **45** with which the scalable monitoring platform is intended to be compatible.

(51) Referring to FIGS. **3A-B** and FIGS. **2A-B**, in one example, the protocol compatibility submodule **44P** allows the request module **44** to generate a target request **52** to retrieve a requested data item **55R** via SNMP. After identifying SNMP as the network management protocol **45**, the protocol compatibility submodule **44P** may select the appropriate management command from within the SNMP protocol, and then identify the device network address **36A** of the managed device **24** and the device level identifier **36D** of the requested data item **55R** using the MIB file **28B**. The request module **44** may then use the identified address data **36** to produce the SNMP management command to retrieve the requested data item **55R** from the appropriate managed device **24**. The targeted request **52** is then transmitted to the target network **20** for delivery to the managed device **24**.

(52) Turning to FIGS. **4** and **5** while also referring to FIG. **1B**, FIGS. **2A-B** and FIGS. **3A-B**, each managed device **24** which receives a targeted request **52** will reply by transmitting a response **70** containing response data **74**. The response data **74** corresponds to the requested data item **55R** of the targeted response **52**, and embodies management data **58** which may be host data **58A** or service data **58B**. In one embodiment, the response **70** is transmitted to the monitoring device **14** via the network management device **22**, whereupon the response data **74** is relayed to the remote monitoring server **12** for evaluation by the data evaluation module **46**. In certain embodiments, the data evaluation module **46** may be implemented on the monitoring device **14** as well as on the remote monitoring server **12**, thus allowing for the response data **74** to be evaluated locally at the network site, thus avoiding delays in evaluation which may be incurred during transmission of the response data **74** to the remote monitoring server **12** via the internet **500**.

(53) The data evaluation module **46** performs a data evaluation function **46E** using evaluation criteria **38E** specified by the monitoring policy **50** of the managed device **24**. The value of the response data is evaluated to determine whether the value of the data represents a change to the functional condition or service task condition of the managed device **24**, by comparing the value against the standard range **38ES** within the evaluation criteria **38E**, as well as any other ranges which indicate the presence or likelihood of an adverse condition, such as the warning range **38EW** or critical range **38EC**.

(54) In certain embodiments, the response data **74** comprises raw response data **74R** which must be converted before evaluation can occur. For example, the managed device **24** may be configured to generate management data in the form of octet strings, which cannot be directly compared against evaluation criteria **38E** expressed using ranges of integer values. The data evaluation module **46** is therefore adapted to perform a data conversion function **46C** to convert the raw response data **74R** into a standardized format which enables direct comparison against the evaluation criteria **38E**. In certain embodiments, the evaluation criteria **38E** may contain data conversion information **38EV** which identifies the correct data type to which the raw response data **74R** will be converted. The data conversion function **46C** may therefore produce converted response data **74C** which is standardized for use with protocol-agnostic monitoring policies **50**.

(55) Turning to FIG. **6** and FIGS. **4-5**, while also referring to FIG. **1B**, FIG. **2B**, and FIGS. **3A-B**, an exemplary data evaluation process **600** is shown. Note that the steps within the exemplary process may be reordered or removed and additional steps may be included, while remaining consistent with the principles set forth within the present disclosure.

(56) At step **602**, a response **70** from one of the managed devices **24** is received by the data evaluation module **46**. Where necessary, the raw response data **74R** is converted to produce converted response data **74C**. Next, at step **604**, the data evaluation module **46** determines whether the management data **58** corresponds to host data **58A** or service data **58B** in order to categorize the response **70** by output type **76**. In one embodiment, the monitoring policy **50** of the managed device **24** contains classifications which identify the output type **76** of each associated data item **38**

of the managed device **24**, and indicate whether the associated data item **38** corresponds to host data **58A** or service data **58B**. Furthermore, the output type **76** may also indicate whether the data is treated as static data **76S** or performative data **76P**.

(57) If the response **70** contains host data **58A**, the data evaluation module **46** performs a host threshold evaluation at step **606** to assess the functional condition of the managed device **24**. In one embodiment, the host threshold evaluation is performed using a threshold comparison in which the value of the host data **58A** is compared against one or more threshold values **38ET**. Threshold values **38ET** may be configured in various ways depending on the managed device **24**, and each threshold value **38ET** may be used to indicate whether the value of the management data **58** signifies a change in the functional condition of the managed device. Each managed device **24** may be assigned a host condition corresponding to an operational host condition or an inoperable host condition. In an example, the host data **58A** item may correspond to a power amplification measurement in which a data value falling below the threshold value **38ET** indicates that the managed device **24** is producing insufficient power to function. As such, the data evaluation module **46** may determine that the managed device **24** is in the inoperable host condition at step **610**, and proceed to generate a host condition error alert at step **612**. Conversely, if the host threshold evaluation at step **606** determines that the value of the host data **58A** is above the threshold value **38ET**, the data evaluation module **46** determines that the managed device **24** is in the operational host condition at step **608**.

(58) In certain embodiments, the data evaluation module **46** may be configured to detect whether an expected response **70** to a targeted request **52** formed as a host status request **55TA** has not been received. In one embodiment, the data evaluation module **46** may determine that a response failure has occurred after a period of time corresponding to a response failure interval has passed. The data evaluation module **46** may proceed with the data evaluation function **46E** on the assumption that the managed device is in the inoperable host condition. In certain embodiments, the host condition of any nodes **26** of the target network which are positioned upstream of the managed device **24** may be evaluated to determine if any of the intermediate nodes **26N** have failed, thus preventing communication with the managed device **24**.

(59) Returning to step **604**, if the management data **58** corresponds to service data **58B**, the data evaluation module **46** will determine if the output type **76** indicates the data is static data **76S** or performative data **76P** at step **614**. Service data **58B** which is static data **76S** does not require evaluation, and may be simply forwarded to the platform data store **30** for storage within the data log **78** at step **616**. One example of static data **76S** may include a serial number assigned to the managed device **24** which may be used to for inventory control purposes. Other examples of static data **76S** include physical location, firmware information, and other types information of descriptive or informative value.

(60) If the service data **58B** corresponds to performative data **76P**, the data evaluation module **46** will perform a service threshold evaluation at step **618** using the evaluation criteria **38E** linked to the associated data item **38** contained in the response **70**. The service threshold evaluation allows the data evaluation module **46** to determine whether the service data **58B** indicates if the condition of the service task is within normal levels, or if an adverse condition exists. In a preferred embodiment, each item of service data **58B** is evaluated against the standard range **38ES**, the warning range **38EW**, and the critical range **38EC**.

(61) The data evaluation module **46** compares the value of the service data **58B** to each of the service ranges at step **618** as part of the service threshold evaluation. If the value of the service data **58B** falls within the standard range **38ES**, the data evaluation module **46** will determine that no adverse service task condition is present at step **620**. No change to the operating mode of the monitoring policy is required, and the monitoring policy **50** continues to operate in the standard operating mode **50S**. The value of the service data **58B** will also be logged with the appropriate timing data for storage within the data log **78**.

(62) If the value of the service data **58B** falls within the warning range **38EW**, the data evaluation module **46** will determine that a warning condition is present at step **624**, and the monitoring policy **50** for the managed device **50** will begin operating in the warning operating mode **50W** at step **626**. In one embodiment, the warning range **38EW** identifies data values of the service data **58B** which indicate an intermediate level of risk or adverse impact.

(63) If the value of the service data **58B** falls within the critical range **38EC** at step **618**, the data evaluation module **46** will determine that a critical condition is present at step **628**, and initiate the critical monitoring mode **50C** of the monitoring policy **50** for the managed device **24** at step **630**. The critical range **38EC** may identify data values of the service data **58B** which indicate a high level of risk or adverse impact.

(64) The data evaluation module **46** may also be configured to detect changes in management data **58** which constitute patterns or trends. For example, the data evaluation module **46** may be configured to detect if the values of service data **58B** returned by a managed device **24** approach an adverse range at recurring times.

(65) The data evaluation module **46** may also be configured to detect degradation in the management data **58** and initiate a change to an adverse operating mode of the monitoring policy **50**. For example, a managed device **24** may return service data **58B** with values which fall within the standard range **38ES** but which steadily increase or decline towards the values constituting the warning range **38EW**. As a result, the data evaluation module **46** may initiate the warning operating mode **50W** for the managed device **24** before the service data **58B** actually falls within the warning range **38EW**.

(66) In one embodiment, the warning operating mode **50W** of a monitoring policy **50** may cause the request module **44** to increase the quantity and frequency of targeted requests **52** sent to any managed device **22** subject to a warning condition, such as by adjusting the request interval **62** within the request parameters **54**. If the service data **58B** represents an erroneous reading, increasing the frequency of requests may provide rapid confirmation regarding whether the adverse condition exists.

(67) In one embodiment, a change in operating mode from the standard operating mode **50S** to one of the adverse operating modes may cause the request module **44** to initiate a confirmation procedure in which targeted requests **52** are transmitted to the managed device **24** as part of a sequence of confirmation requests. If the responses **70** to the confirmation requests indicate that the service data **58B** values have returned to the standard range **38ES**, the data evaluation module **46** may determine that the managed device **24** is operating normally and reinitiate the standard operating mode **50S** of the monitoring policy **50**. In one example, the request interval **62** may be increased from five minutes to an interval of three minutes, and the sequence of confirmation requests may comprise three targeted requests **52**.

(68) Turning to FIG. 7 while also referring to FIGS. 4-6 and FIG. 1A, the adverse operating modes of a monitoring policy **50** may also define management actions **102** to be initiated in response to changes in functional condition or service task conditions. The management actions **102** may include alerting users of the changes, or initiation of actions or steps to mitigate or avert adverse outcomes.

(69) Management actions **102** may include the notification module **90** notifying users via platform alerts **94** displayed on the platform interface **18**. For example, the notification module **90** may generate a platform alert **94** upon each detection of an adverse condition by the data evaluation module **46**. The platform alert **94** may contain information which identifies the managed device **24** which is subject to the adverse condition. The platform alert **94** may also include data from the data log **78** which identifies and describes the adverse condition and further provides visualization of the relevant management data **58**. The platform alert **94** may also provide static service data **58B** such as serial number, firmware information, physical location data, or other types of data or information as required. The notification module **90** may also forward the content of platform alerts **94** using

email, SMS, or a messaging or communication platform.

(70) In one embodiment, as part of a management action **102**, the remote monitoring server **12** or monitoring device **14** may cause a management command to be sent to the managed device **24** which instructs the managed device to undertake a managed device action. The managed device action may correspond to an action or function which is supported by the network management protocol **45** in use by the target network **20** and the managed device **24**. For example, the management action **102** may instruct the managed device **24** to undergo a reset, reboot, or shut down sequence in order to mitigate an adverse condition. The management command may be relayed through the network management device **22** or be sent directly from the monitoring device **14** to the managed device **24** in accordance with the capabilities of the target network **20** and the network management protocol **45**.

(71) In certain embodiments, the management action **102** may cause the notification module **90** to transmit an external action command **98** to an external management system **100**. The external management system **100** may be a system or platform with which the remote monitoring platform **10** can communicate via a unified operating framework **104** which provides interoperability between systems and a common automation platform for the execution of network functions and services. The unified operating framework **104** allows the external management system **100** to interpret and execute the external action command **98**. An example of a unified operating framework **104** is the Open Network Automation Platform (ONAP).

(72) In one embodiment, the external management system **100** controls one or more devices which are capable of carrying out automated actions which can mitigate the adverse condition. The external action command **98** may include external action parameters **98P** which govern the execution of the automated action. For example, the external management system **100** may be a drone control system capable of receiving commands through the unified operation framework **104**. In the present example, the monitored device **24** may be a moisture sensor, and the external action parameters **98P** may be used to specify the physical location within the network site which is monitored by the managed device **24**. In reaction to service data **58B** from the managed device **24** indicating dangerously dry conditions at the physical location, the remote monitoring platform **10** causes the notification module **90** to transmit an external action command **98** to the drone controller which identifies the physical location. The drone control system may then execute the external action command **98** by directing a flight capable drone equipped with water dispensing equipment to fly to the physical location specified in the external action parameters **98P**, and dispense water to mitigate the adverse condition. Note that the example provided above regarding the use of drones is merely illustrative and is not intended to limit the applicability or functionality of external action commands **98**.

(73) In certain embodiments, the monitoring policies **50** may initiate management actions **102** if confirmation requests initiated as part of a confirmation procedure indicate that the adverse condition persists. Furthermore, the monitoring policies **50** may specify a sequence of management actions **102** which are carried out if the adverse condition persists following the preceding management action **102** within the sequence. For example, a monitoring policy **50** may first cause a management command **102** to be sent which resets a managed device **24**, followed by a platform alert **94** to initiate a maintenance request if the adverse condition persists following the resetting of the managed device **24**.

(74) As will be appreciated by one skilled in the art, aspects of the present disclosure may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having

computer readable program code embodied thereon.

(75) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium (including, but not limited to, non-transitory computer readable storage media). A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus or device.

(76) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate or transport a program for use by or in connection with an instruction execution system, apparatus or device.

(77) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(78) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. Other types of languages include XML, XBRL and HTML5. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(79) Aspects of the present disclosure are described below with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the disclosure. Each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(80) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the

flowchart and/or block diagram block or blocks.

(81) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(82) The flowchart and block diagrams in the Figures illustrate the architecture, functionality and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(83) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(84) The flow diagrams depicted herein are just one example. There may be many variations to this diagram or the steps (or operations) described therein without departing from the spirit of the disclosure. For instance, the steps may be performed in a differing order and/or steps may be added, deleted and/or modified. All of these variations are considered a part of the claimed disclosure.

(85) In conclusion, herein is presented a scalable monitoring platform. The disclosure is illustrated by example in the drawing figures, and throughout the written description. It should be understood that numerous variations are possible, while adhering to the inventive concept. Such variations are contemplated as being a part of the present disclosure.

Claims

1. A network monitoring system for monitoring a plurality of managed devices in a target network, comprising: a remote monitoring server; a monitoring device configured to communicate with a target network comprising a network management device and a plurality of managed devices, the monitoring device is co-located with the target network, and the remote monitoring server is configured to communicate with the monitoring device, wherein each of the plurality of managed devices stores management data comprising at least one of host data describing a functional condition of the managed device and service data describing a service task which the managed device is configured to carry out, the monitoring device comprising: a processor; and a non-transitory computer-readable storage media storing code, the code being executable by the

processor to perform operations comprising: separately dynamically generating a plurality of targeted requests for management data corresponding to each of the plurality of managed devices; separately directly transmitting each of the plurality of targeted requests to the corresponding one of the plurality of managed devices via the network management device; and separately receiving, from each of the plurality of managed devices, the management data corresponding to the one of the plurality of managed devices; wherein the remote monitoring server is configured to identify a network management protocol used by the target network and cause the monitoring device to format each of the plurality of targeted requests to include network management commands native to the network management protocol; the remote monitoring server is further configured to retrieve at least one of configuration data corresponding to the target network, the management data corresponding to each of the plurality of managed devices, and address data which identifies each of the plurality of managed devices by a unique address; the remote monitoring server is further configured to locate device level identifiers associated with each of the plurality of targeted requests, wherein each device level identifier is associated with a subcomponent of one of the managed devices which is configured to gather raw service data; the remote monitoring server is further configured to retrieve the raw service data corresponding to one of the plurality of managed devices based on the device level identifier of the managed device, interpret the raw service data, and execute at least one protocol-agnostic monitoring policy with respect to at least one of the plurality of managed devices; the remote monitoring server is configured to detect, based on the management data, an adverse condition associated with one of the plurality of managed devices, and access the monitoring policy to determine at least one adverse operating mode associated with said managed device; the remote monitoring server is further configured to execute the at least one adverse operating mode and generate additional targeted requests for management data to said managed device at a predetermined frequency to verify the adverse condition; and the remote monitoring server is further configured to initiate at least one management action in response to verification of the adverse condition, the at least one management action comprising notifying a user of the adverse condition.

2. The network monitoring system of claim 1, wherein separately directly transmitting each of the plurality of targeted requests comprises transmitting each of the plurality of targeted requests in parallel.

3. A method for monitoring a plurality of managed devices in a target network having a network management device, comprising: storing, by each of a plurality of managed devices within a target network, management data comprising at least one of host data associated with a functional condition of the one of the plurality of managed devices and service data associated with a service task condition of the one of the plurality of managed devices; co-locating a monitoring device with the target network; providing a remote monitoring server, wherein the remote monitoring server is configured to communicate remotely with at least one of the monitoring device and the network management device; storing, by at least one of the remote monitoring server and the monitoring device, a monitoring policy associated with each of the plurality of managed devices, wherein the monitoring policy comprises at least one rule for evaluating the management data; separately dynamically generating a plurality of targeted requests for the management data corresponding to each of the plurality of managed devices, wherein each of the plurality of targeted requests corresponds to exactly one of the plurality of managed devices; separately directly transmitting, by the monitoring device, each of the plurality of targeted requests to the corresponding one of the plurality of managed devices via the network management device; separately receiving, in response to each of the plurality of targeted requests, the management data corresponding to each of the plurality of managed devices; detecting, based on the management data, an adverse condition associated with the corresponding one of the plurality of managed devices; accessing the monitoring policy to determine at least one adverse operating mode associated with the one of the plurality of managed devices; executing the at least one adverse operating mode; generating, in

response to the at least one adverse operating mode, additional targeted requests for management data; transmitting, to the one of the plurality of managed devices, the additional targeted requests at a predetermined increased frequency to verify the adverse condition; and initiating, in response to verification of the adverse condition, at least one management action comprising notifying a user of the adverse condition.

4. The method of claim 3, further comprising identifying, via the remote monitoring server, a network management protocol used by the target network, and configuring each of the plurality of targeted requests to include network management commands native to the network management protocol.

5. The method of claim 4, further comprising: locating, via the remote monitoring server, device level identifiers associated with each of the plurality of targeted requests, wherein each of the device level identifiers is associated with a subcomponent configured to gather raw service data; and retrieving raw service data based thereon.

6. The method of claim 5, further comprising interpreting the raw service data to execute at least one protocol-agnostic monitoring policy with respect to at least one of the plurality of managed devices.

7. The method of claim 6, wherein detecting the adverse condition further comprises: evaluating the management data to detect a change in at least one of the functional condition and the service task condition of the one of the plurality of managed devices; identifying, based on the change, the adverse condition associated with the one of the plurality of managed devices; and comparing the management data corresponding to the one of the plurality of managed devices to a predetermined data range to assess a severity of the adverse condition.

8. The method of claim 3, wherein separately transmitting each of the targeted requests comprises dynamically scheduling transmission of each of the targeted requests to minimize a risk of the adverse condition.
